Engel's Law Holds for Groceries, Not for Restaurants

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Question

Engel (1857) observed in Saxon household budgets that the poorer a family, the larger the share of its spending that goes to food. In modern terms, food has an income elasticity below one: when total spending rises by one percent, food spending rises by less. Does Engel's law still hold in recent US data, and does it hold for all food or only for some of it?

Result

It holds in every year from 1984 to 2024. Across the five income groups of the Consumer Expenditure Survey, food spending rises by 0.77% for each 1% of total spending on average, and the elasticity is below one in all 41 years. The law is driven entirely by groceries, whose elasticity averages 0.58. Spending on food away from home has an elasticity of 1.07, above one in 40 of 41 years: in 2024 it took between 4.7% and 5.2% of total spending in every income group, while groceries fell from 11.0% of spending in the lowest group to 6.2% in the highest. For the average household the food share fell from 14.7% in 1984–1993 to 12.8% in 2005–2019 and has not fallen since. Figure 1 shows the three patterns.

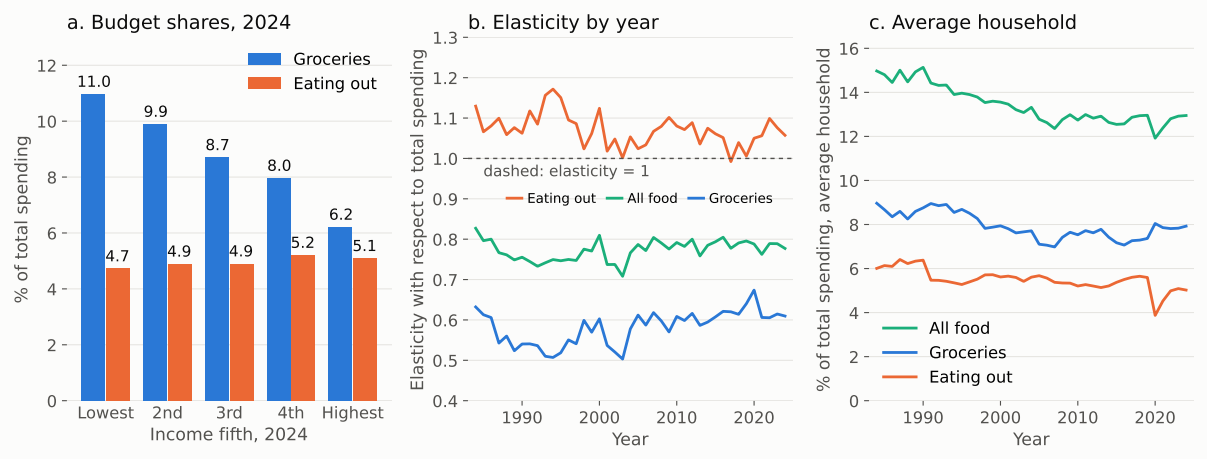


Figure 1: Engel's law in the US Consumer Expenditure Survey, 1984–2024. Panel a: spending on groceries (food at home) and on food away from home as a percentage of total spending, by income fifth, 2024. Panel b: elasticity of spending with respect to total spending across the five income groups, by year; the dashed line marks an elasticity of one. Panel c: food spending as a percentage of total spending for all consumer units.

Why it matters

Engel's law is built into how the United States measures poverty. In the early 1960s Mollie Orshansky set the official poverty thresholds at three times the cost of a minimal food plan, because the 1955 Household Food Consumption Survey showed families spending about a third of their after-tax income on food (Orshansky, 1965). The thresholds are still updated from that base for price inflation only. In 2024 food took 15.7% of total spending in the lowest income fifth and 12.9% for the average household; applying Orshansky's reasoning to these shares would give a multiplier of about 6 to 8 rather than 3. These shares are of spending rather than after-tax income, so the comparison is indicative, but the direction is the

one Engel's law predicts as incomes rise. Engel curves are also used to estimate bias in consumer price indexes (Hamilton, 2001; Costa, 2001), which relies on a stable relation between the food share and real income. The split shown here matters for that use: the relation is stable for groceries, while spending on eating out rises at least in proportion to the budget.

Data and design

Data. Bureau of Labor Statistics, Consumer Expenditure Survey, published annual means for all consumer units and for each fifth of income before taxes, 1984–2024, obtained through the BLS public data interface. Items are food (total), food at home, food away from home, total expenditures and income before taxes. A consumer unit is roughly a household.

Shares. Each item is divided by total expenditures of the same group and year.

Elasticity. For each year, the elasticity is the least-squares slope of the log of the item on the log of total expenditures across the five income groups. Engel's law for an item means a slope below one. The elasticity of total food with respect to income before taxes, rather than total spending, averages 0.42, because spending rises less than income across the groups.

	1984–1993	1994–2004	2005–2019	2020–2024
Elasticity, all food	0.77	0.75	0.79	0.78
Elasticity, groceries	0.56	0.55	0.61	0.62
Elasticity, eating out	1.09	1.08	1.05	1.07
Food share, lowest income fifth (%)	17.6	16.7	15.7	15.4
Food share, highest income fifth (%)	12.7	11.7	11.3	11.1
Food share, all consumer units (%)	14.7	13.6	12.8	12.6
of which groceries (%)	8.7	8.0	7.4	7.9
of which eating out (%)	6.0	5.5	5.4	4.7

Table 1: Engel's law by period. Elasticities are averages of the yearly cross-group slopes; shares are averages of yearly shares of total expenditures. The 2020–2024 averages include the 2020 pandemic year, when eating out fell to 3.9% of spending.

The food share falls from the lowest to the highest income group in every period, and the yearly decline is monotone across all five groups in 34 of 41 years. In the other seven years one adjacent pair is reversed: in 1984, 1986, 1991 and 2000 the lowest fifth spends a slightly smaller share on food than the second fifth, and in 2015, 2018 and 2022 two middle groups are reversed. The grocery elasticity ranges from 0.50 to 0.67 across years and the eating-out elasticity from 0.99 to 1.17. The elasticities show no clear trend; the grocery elasticity is somewhat higher after 2005 than before. The decline in the average food share came from both components until the mid-2000s; since then the share has stayed between about 12.4% and 13.0% outside 2020.

Caveats

The estimates use five group means per year, not individual households, so they describe the pattern across income groups rather than the response of any one household to a change in its income. Higher income groups have larger households; shares are unaffected by household size, but elasticities computed from household totals partly reflect it. Income groups are defined by income before taxes, which is noisy for some households, such as retirees spending savings. Food away from home includes spending while travelling and on meals at work.

Relation to the literature

Houthakker (1957) confirmed Engel's law across many countries a century after Engel, and Chai and Moneta (2010) review its history. That spending on food away from home behaves differently from groceries is known in food economics. This note is a clean update on 1984–2024 published US data, with the split by component and the link to the poverty line made explicit; it does not claim a new finding.

References

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Data: US Bureau of Labor Statistics, Consumer Expenditure Survey (public). Drafted with AI assistance and checked before publication.